View Reviews

Paper ID
816

Paper Title
Reproducible 6-DOF Wrist-Roll Retrofit for OpenMANIPULATOR-X

Reviewer #1

Questions

5. Detailed Comments and Suggestions for improving the paper.
The paper presents a relevant and practically useful contribution by introducing a low-cost, reproducible wrist-roll retrofit that upgrades the OpenMANIPULATOR-X from a 5-DOF to a 6-DOF robotic manipulator. The integration of mechanical design, ROS 2/MoveIt 2 configuration, pose-based evaluation, statistical analysis, and physical hardware validation is a strength of the work. The reported improvement in orientation reachability and the discussion of simulation-to-real discrepancies make the study potentially valuable for robotics education and research. However, the following points should be addressed to strengthen the technical rigor and clarity of the manuscript.

1. **Clarify the fairness of the planning comparison:** The corridor-planning benchmark uses different orientation-reachability capabilities for the two configurations. Since the 5-DOF arm cannot achieve the non-zero-roll targets at the fixed position, the reported planning-success difference combines kinematic feasibility and motion-planning performance. The authors should distinguish these effects more explicitly and, where possible, report results on a common set of targets reachable by both configurations.

2. **Strengthen the statistical analysis:** The narrowest-corridor comparison uses repeated planning trials across seven goals. The authors should clarify whether the repeated trials are statistically independent and account for potential dependence among trials associated with the same goal. A goal-wise or otherwise appropriately structured analysis would help establish the robustness of the reported significance. Confidence intervals and effect sizes should be reported consistently wherever applicable.

3. **Expand the real-hardware validation:** The physical evaluation is limited to one retrofitted prototype. The reported strict Cartesian position success rate of 40% within 5 mm, compared with 97.5% orientation success within 5 degrees, deserves further discussion. The authors should clarify the practical implications of the positional error and

explain whether calibration, backlash compensation, trajectory tuning, or mechanical improvements could reduce it. Validation on additional units would strengthen the reproducibility claims.

4. **Improve figure readability:** Ensure that all figures have sufficiently large labels, readable legends, clearly marked axes, and consistent notation. Where appropriate, include uncertainty indicators or confidence intervals in performance plots. The figures should remain legible when printed in the conference's two-column format.

5. **Improve reproducibility details:** The paper refers to a project repository for the URDF/Xacro modifications, controller configurations, solver tolerances, and experimental settings. Ensure that the final submission or publication provides accessible artifacts, a clear setup procedure, dependency versions, and instructions for reproducing the reported experiments.

6. **Clarify the interpretation of the kinematic compatibility results:** The reported zero overlap between the sampled 6-DOF poses and the 5-DOF arm's solvable poses is strongly influenced by the sampling strategy and strict IK tolerances. The authors acknowledge this limitation, but the distinction between a sampling-dependent result and a general property of the manipulator should remain explicit throughout the results and conclusion.

7. **Discuss the planning results more cautiously:** The proposed explanation that higher wrist-roll angles improve corridor passage by changing the gripper's effective cross-section is plausible, but the paper acknowledges that this mechanism was not directly measured. The authors should retain this as a hypothesis and consider adding geometric or collision-clearance measurements in future work.

8. **Review presentation and references:** Check the consistency of technical terminology, mathematical notation, units, figure and table references, and IEEE reference formatting. Ensure that all citations and bibliographic details are accurate and that the final manuscript complies with the conference's official template.

Overall, the work has practical merit and a promising experimental foundation. Addressing the comparison methodology, statistical treatment, and hardware-performance limitations would improve the reliability and impact of the contribution.